

Experiment:- study of LAN Topology, Network devices and Address configuration.

Introduction:-

A Local Area Network is a network that connects computer and devices within a limited geographical area such as a home, school, office or college library. LAN enables communication, data sharing, internet access, and resources sharing among connected devices.

Objectives :-

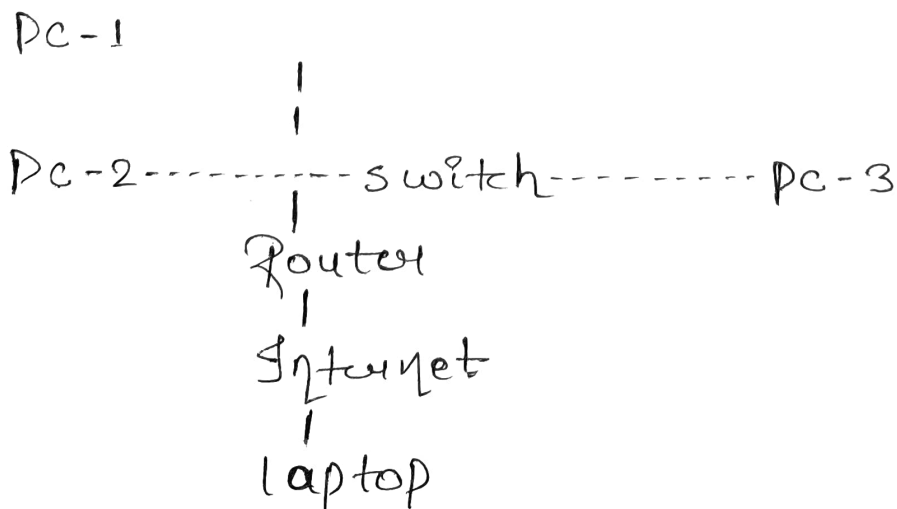
- 1) To study the LAN topology used in college.
- 2) To identify networking devices.
- 3) To analyze wired and wireless connections.
- 4) To measure network speed
- 5) To obtain MAC address, IP address, and subnet mask.

2. LAN Topology:-

Star topology

In this experiment, star topology is used. In a star topology, all devices are connected to a central switch or router.

Diagram



## Advantages of Star Topology:-

- Easy to manage
- Fast communication
- Failure of one device does not effect others
- Easy trouble shooting.

## • Network Devices

| <u>Device</u> | <u>Function</u> |
|---------------|-----------------|
|---------------|-----------------|

|        |                               |
|--------|-------------------------------|
| Router | → connect LAN to the Internet |
|--------|-------------------------------|

|        |                                    |
|--------|------------------------------------|
| Switch | → connects multiple devices in LAN |
|--------|------------------------------------|

|              |                                  |
|--------------|----------------------------------|
| Access Point | → provides wireless connectivity |
|--------------|----------------------------------|

|          |                          |
|----------|--------------------------|
| Computer | → send and receives data |
|----------|--------------------------|

|                |                                |
|----------------|--------------------------------|
| Ethernet cable | → used for wired communication |
|----------------|--------------------------------|

## Network topology

The campus network follows a hybrid Hierarchical topology combining.

- Tree Topology
- Extended star topology
- Ring topology

→ The network hierarchical layer

① Core layer

② Distribution layer

③ Access layer

→ core layer

\* core layer contain server room and server room have firewall, Router, 2-13+ switches, Patch Rack, Server, inverter, etc.

→ Distribution layer:-

\* Distribution layer contain 4 - switches.

\* The four switches are →

① Main Build - R7 (Room no - 211, )

② Main Build - R1 (Room no - 301)

③ Main Build - R6 (Room no - 202)

④ Main Build - R2 (Room no 201)

\* This 4-switches contains →

① 18 Port Gigabit Easy smart switch with 16-port.

TL-SG1218MPE

② SG3428XMP → 24 port Gigabit & 4 port 10Gbps, L2+ Managed switches with 24 port PoE+.

Now let's discuss these 2 above components →

① TP-Link TL-SG1218MPE

\* Likely used for: CCTV camera, wifi access point.

\* 18 port Gigabit "Easy Smart" switch.

\* 16 port support PoE+

→ Access Layer

Access layer contain 3-switches i.e.

1) Main Build R1 [Room no - 301]

2) Main Build R4 [Room no - 05]

3) Main Build R5 [Room no - 16]

Every switches act as access layer which helps to provide network connectivity to all classroom, lab Faculty Room etc.

Access layer directly connect:

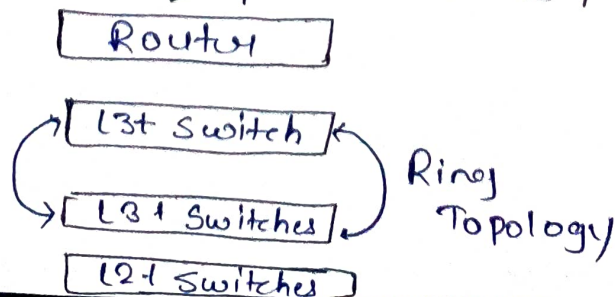
① Ethernet port

② Wireless Ap

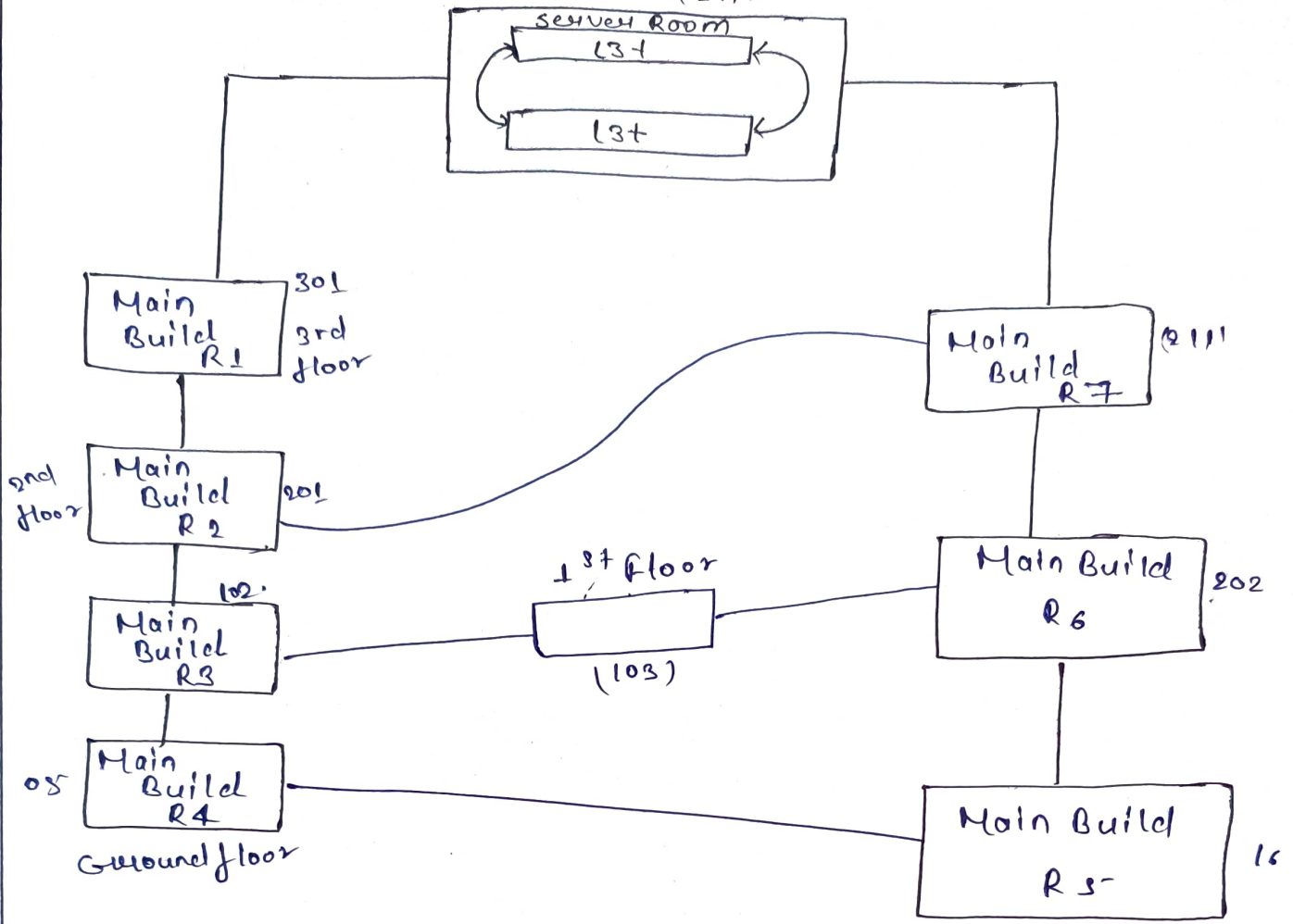
③ Lab system

④ outdoor wireless Ap.

→ Server Room Architecture / Topology.



## Topology (211)



- R1, R7, R3, R6 → Distribution Layer
- Server Room → Core layer
- R1, R2, R3, R4, R5, R6, R7 → Access layer.

## System Network Details

| Parameter                           | Value                        |
|-------------------------------------|------------------------------|
| <del>Router</del> Wifi Name (SSID)  | Motorola edge 20 fusion-7768 |
| <del>core switch</del> IPv4 Address | 192.168.29.73                |
| Subnet Mask                         | 255.255.255.0                |
| Default Gateway                     | 192.168.29.1                 |
| MAC Address                         | 28-DO-43-36-AD-CB            |

## Wireless connection & speed

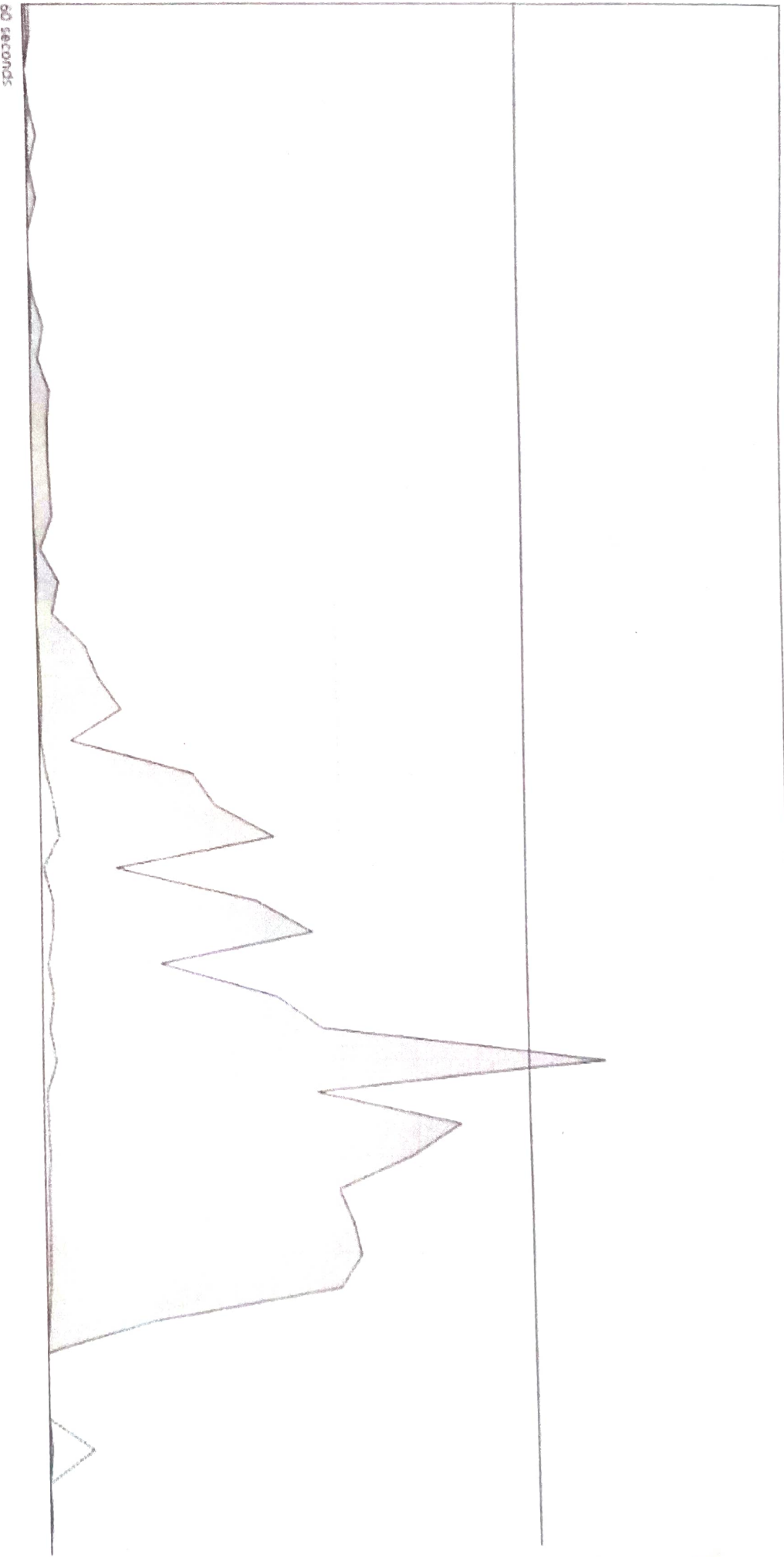
The Laptop was connected through wireless communication using wi-fi. The observed receive speed during the experiment was approximately 59.3 MBps

# WIRELESS SPEED

Wi-Fi

MediaTek MT792

Throughput



Send

Adapter name:

Wi-Fi

0 Kbps

SSID:

motorola edge 20 fusion\_7768

Connection type:

802.11n

Receive

IPv4 address:

10.134.115.233

IPv6 address:

2409:40e4:22:29fc:e4ce18a0:76f60a8

0 Kbps

Signal strength:



| connection type     | Speed     |
|---------------------|-----------|
| wireless connection | 59.3 Mbps |
| Network Band        | 2.4 GHz   |

Differentiate Between wired & wireless Networks

| wired Network                                                                                                                                                                                    | wireless Network                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>* Faster communication</li> <li>* Stable connection</li> <li>* Low Latency</li> <li>* uses ethernet cable</li> <li>* Dedicated Physical Medium</li> </ul> | <ul style="list-style-type: none"> <li>* Flexible communication</li> <li>* Affected by signal strength</li> <li>* Higher Latency</li> <li>* uses radio waves</li> <li>* shared Radio Medium</li> </ul> |

Q Role of Subnet mask in network?

Ans:- A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

\* It is important for:

- ① Network Segment
- ② Effective routing
- ③ communication b/w devices.

→ OBSERVATION

During the study of the college network infrastructure:

- ① A centralized server room was used as network backbone.
- ② two L3 switches were connected in redundant cyclic structure.
- ③ Managed TP link PoE switches were used in network cabinets.

→ Advantages of Existing Networks?

- \* Scalable ~~and~~ architecture
- \* Centralized management
- \* Support PoE devices
- \* Easy expansion of floors & classroom
- \* Support both wireless & wired.
- \* Redundant link for failover.

### Conclusion

The observed campus LAN follow a Hybrid Hierarchical topology consist of Tree, Extended Star, Ring. The network uses managed PoE switches wireless access point, Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.